



JULIAN CABRERA MARCEGLIA

Manufacturing Systems Engineer | Industrial Data Platforms | Manufacturing Information Systems | Digital Manufacturing

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PROFESSIONAL SUMMARY

Manufacturing Systems Engineer with 21+ years of experience in automotive manufacturing environments, combining manufacturing operations expertise, quality systems knowledge and industrial data management capabilities.

Experienced in supporting Toyota manufacturing programs through roles spanning production, quality engineering support and manufacturing information systems. Specialized in identifying manufacturing information challenges and developing digital solutions that improve traceability, operational visibility and engineering decision-making.

Designed and implemented a centralized Manufacturing Measurement Data Platform integrating multiple industrial data sources and managing more than 50,000 manufacturing records. Improved reporting efficiency approximately 10x while creating scalable information architecture supporting manufacturing operations and engineering teams.

Combines extensive shop-floor experience with software engineering and database development capabilities to support manufacturing systems, industrial data solutions and digital transformation initiatives.

CORE EXPERTISE

Manufacturing Systems

- Manufacturing Information Systems
- Manufacturing Data Management
- Shop Floor Systems
- Manufacturing Process Support
- Digital Manufacturing

- Manufacturing Traceability
- Production Information Management
- Operational Visibility

Industrial Data & Analytics

- Industrial Data Platforms
- SQL
- Database Design
- Data Modeling
- Reporting Automation
- Manufacturing Analytics
- Statistical Process Control
- Engineering Data Management

Manufacturing Operations

- Automotive Manufacturing
- Quality Systems
- Continuous Improvement
- Root Cause Analysis
- KPI Management
- Cross-Functional Collaboration
- Process Capability Analysis
- Manufacturing Problem Solving

Software & Technical

- Python
- SQL
- DuckDB
- Pandas
- Database Applications
- Data Processing
- Reporting Systems
- Software Architecture

PROFESSIONAL EXPERIENCE

Metalsa

Quality Technician Supervisor

Sep 2016 – Present

Buenos Aires Province, Argentina

Support Toyota automotive manufacturing programs by combining manufacturing process knowledge, quality systems expertise and digital information solutions.

Work directly with engineering, production and quality teams to improve access to manufacturing information and support operational decision-making.

Manufacturing Systems & Data Solutions

- Designed and implemented a centralized Manufacturing Measurement Data Platform integrating multiple industrial measurement sources.
- Integrated CMM, FARO and manual inspection information into a unified engineering information environment.
- Managed more than 50,000 manufacturing measurement records.
- Automated engineering reporting processes, improving efficiency approximately 10x.
- Improved manufacturing traceability and information accessibility.
- Developed scalable information management solutions supporting increasing manufacturing complexity.
- Supported manufacturing stakeholders through data-driven analysis and reporting solutions.

Manufacturing Quality & Operations Support

- Lead KPI review meetings and continuous improvement activities.
- Support manufacturing teams through SPC, MSA, GR&R and process capability analysis.
- Participate in root-cause investigations and corrective action initiatives.
- Develop and maintain measurement programs using PC-DMIS and FARO systems.

- Collaborate with production, engineering and quality stakeholders.
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Quality Technician

Sep 2008 – Sep 2016

Supported automotive manufacturing quality operations for Toyota programs.

- Performed dimensional analysis and manufacturing system verification activities.
 - Generated and analyzed manufacturing information from measurement systems.
 - Developed and maintained hundreds of inspection programs.
 - Conducted statistical analysis supporting manufacturing decisions.
 - Supported engineering teams in process improvement initiatives.
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Production Welder

Sep 2004 – Sep 2008

- Performed automotive production operations within high-volume manufacturing environments.
- Developed practical understanding of production systems, manufacturing constraints and shop-floor operations.

SELECTED DIGITAL TRANSFORMATION PROJECTS

Manufacturing Measurement Data Platform

Manufacturing Information System | Industrial Data Integration

Designed and implemented a digital manufacturing solution to improve production information management.

Manufacturing Challenge

Manufacturing information was generated from different sources, creating challenges related to:

- Data accessibility.
- Traceability.
- Reporting efficiency.
- Engineering analysis.

Solution

Developed a centralized platform integrating:

- CMM measurement data.
- FARO inspection information.
- Manual inspection records.
- Structured manufacturing datasets.
- Automated reporting workflows.

Impact

- Centralized 50,000+ manufacturing records.
- Improved manufacturing information visibility.
- Reduced reporting effort approximately 10x.
- Enabled scalable engineering analysis.

Real-Time Quality Prioritization System

Manufacturing Analytics | Quality Decision Support

Developed a manufacturing analytics solution to transform quality information into prioritized engineering actions.

Capabilities

- Automated data processing.
- Manufacturing performance analysis.
- Quality risk identification.
- Engineering decision support.

Impact

- Improved identification of manufacturing issues.
- Reduced manual analysis effort.

- Supported proactive quality improvement.

TECHNICAL CAPABILITIES

Manufacturing Systems	Data and Databases	Manufacturing Engineering	Software Development
• Manufacturing Information Systems • Digital Manufacturing • Manufacturing Data Platforms • Production Information Management • Engineering Information Systems	• SQL • Database Design • Data Modeling • Reporting Automation • Data Analysis	• Automotive Manufacturing • Quality Systems • GD&T • SPC • MSA • GR&R • Process Capability • CMM • FARO • PC-DMIS	• Python • Backend Development • Database Applications • Software Architecture • Node.js • React.js • Vba

PROFESSIONAL POSITIONING

Manufacturing Systems Engineer focused on connecting manufacturing operations, industrial data and digital solutions to improve shop-floor visibility, information management and operational decision-making. Combines extensive automotive manufacturing experience with industrial data capabilities and software engineering skills to support manufacturing systems improvement and digital manufacturing initiatives.